

A PROJECT TITLE

ON

Wine Quality Prediction Using Decision Tree, Random Forest, and Gradient Boosting
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE

Name of Title: Wine Quality Prediction Using Decision Tree, Random Forest, and Gradient Boosting

1. Abstract:

Wine quality prediction is an important application of machine learning that can assist in evaluating wine based on its physicochemical properties. This project focuses on developing a machine learning-based system to predict wine quality using features such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol content. The proposed system implements and compares three supervised machine learning algorithms: Decision Tree, Random Forest, and Gradient Boosting. The wine quality dataset is first preprocessed to ensure that the data is suitable for model training, after which it is divided into training and testing datasets. Each machine learning model is trained using the selected input features and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. The performance of the three algorithms is then compared to identify the model that provides the most accurate and reliable predictions. Decision Tree provides an interpretable approach by creating a sequence of decision rules, while Random Forest improves prediction by combining multiple decision trees. Gradient Boosting further enhances performance by sequentially correcting errors made by previous models. The project demonstrates the practical application of machine learning in quality assessment and provides insights into how different algorithms perform on the same dataset. The developed system can potentially support automated and consistent wine quality evaluation.

2. INTRODUCTION:

Wine quality assessment is an important process in the food and beverage industry, as the quality of wine is influenced by various physicochemical properties such as acidity, alcohol content, pH, sulphates, chlorides, residual sugar, and sulfur dioxide. Traditionally, wine quality is evaluated through laboratory analysis and sensory examination by experts. However, these methods can be time-consuming and may be affected by human judgment. With the advancement of machine learning, data-driven approaches can be used to analyze these properties and predict wine quality more efficiently and consistently.

This project, "Wine Quality Prediction Using Decision Tree, Random Forest, and Gradient Boosting," aims to develop a machine learning system capable of predicting wine quality based on its physicochemical characteristics. Three supervised machine learning algorithms—Decision Tree, Random Forest, and Gradient Boosting—are implemented and compared to analyze their predictive performance. The wine quality dataset is first prepared through appropriate data preprocessing and then divided into training and testing sets. The models are trained using the selected features and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. The results are compared to identify the algorithm that provides the most reliable predictions.

The project demonstrates the practical application of machine learning in solving a real-world classification problem while providing an understanding of data preprocessing, model training, prediction, and performance evaluation. It also highlights the advantages of comparing multiple algorithms to select a suitable model for a given dataset.

3. SOFTWARE REQUIREMENTS:

Software / Tool	Requirement
Operating System	Windows 11
Programming Language	Python
IDE / Development Environment	Google Colab
Machine Learning Library	Scikit-learn
Data Processing	NumPy
Data Visualization	Matplotlib, Seaborn
Model Evaluation	Scikit-learn Metrics
Dataset	Wine Quality Dataset (CSV format)
Web Browser	Google Chrome
Package Manager	pip / Anaconda

Signature of the Guide




Course Instructor